Differential Equations

Questions 1–46

2.40 — Question 1. Solve the following differential equations:

a) $\dfrac{dy}{dx} = y$

b) $\dfrac{dy}{dx} = \dfrac{y}{x}$

c) $\dfrac{dy}{dx} = \dfrac{y}{x^2}$

d) $\dfrac{dy}{dx} = \dfrac{y}{6x^4}$

2.40 — Question 2. Solve the following differential equations:

a) $\dfrac{dy}{dx} = \dfrac{x}{y}$

b) $\dfrac{dy}{dx} = \dfrac{x^2}{y^2}$

c) $\dfrac{dy}{dx} = \dfrac{x^3}{2y}$

d) $\dfrac{dy}{dx} = \dfrac{x^5}{3y^3}$

2.40 — Question 3. Given that $x = 1$ when $t = 1$, solve the differential equation

$$\frac{dx}{dt} = \frac{x^4}{10t}$$

2.40 — Question 4. Given that $x = 4$ when $t = 0$, solve the differential equation

$$\frac{dx}{dt} = \frac{t}{\sqrt{x}}$$

Hence find the value of x when $t = 2\sqrt{39}$.

2.40 — Question 5. Solve the following differential equations:

a) $\frac{dy}{dx} = \frac{y^2}{2x}$

b) $\frac{dy}{dx} = \frac{y^2}{x^2}$

c) $\frac{dy}{dx} = \frac{y^3}{3x^3}$

2.40 — Question 6. Solve the following differential equations:

a) $\frac{dy}{dx} = xy$

b) $\frac{dy}{dx} = x^2y$

c) $\frac{dy}{dx} = 4x^3y$

d) $\frac{dy}{dx} = y\sqrt{x}$

2.40 — Question 7. Given that $P = 500$ when $t = 0$, solve the differential equation

$$\frac{dP}{dt} = -100Pt^3$$

Hence find the value of P when $t = 0.7$, giving your answer to three significant figures.

2.40 — Question 8. Solve the following differential equations:

a) $\frac{dy}{dx} = xy^2$

b) $\frac{dy}{dx} = x^2y^2$

c) $\frac{dy}{dx} = \frac{2}{3}(xy)^3$

2.40 — Question 9. Given that $y(1) = 6$, solve the differential equation

$$\frac{dy}{dx} = \frac{1}{3}(xy)^3$$

Hence find the exact possible values of $y(0)$.

2.40 — Question 10. Solve the following differential equations:

a) $\dfrac{dy}{dx} = \dfrac{\sin x}{y}$

b) $\dfrac{dy}{dx} = \dfrac{\cos x}{4y}$

c) $\dfrac{dy}{dx} = \dfrac{\sin 2x}{3\sqrt{y}}$

d) $\dfrac{dy}{dx} = \dfrac{5\cos 4x}{\sqrt{16y}}$

2.40 — Question 11.

a) Given that $y(0) = 0$, solve the differential equation:

$$\frac{dy}{dx} = \frac{32\cos 3x}{\sqrt{8y}}$$

b) State the exact maximum value of y

2.40 — Question 12. Solve the following differential equations:

a) $\dfrac{dy}{dx} = \dfrac{\cos 2x}{e^y}$

b) $\dfrac{dy}{dx} = \dfrac{y+1}{x+1}$

c) $\dfrac{dy}{dx} = \dfrac{12x^2}{e^{0.2y}}$

d) $\dfrac{dy}{dx} = \dfrac{4y-3}{e^{0.2x}}$

2.40 — Question 13. Solve the following differential equations:

a) $\frac{dy}{dx} = e^{x-y}$

b) $\frac{dy}{dx} = e^{y-x}$

c) $\frac{dy}{dx} = 2e^{4x-3y}$

d) $\frac{dy}{dx} = 8e^{4y-\frac{1}{2}x}$

2.40 — Question 14. Given that $y(0) = 0$, solve the differential equation

$$\frac{dy}{dx} = e^{y-6x}$$

Hence find the exact value of $y(0.5)$.

2.40 — Question 15. Solve the differential equation

$$\frac{24}{y^{1/3}}\frac{dy}{dx} = \frac{4-x}{x}$$

to show that

$$y = \pm\frac{(k - x + 4\ln x)^{3/2}}{216}$$

where $x > 0$ and k is a constant.

2.40 — Question 16. The height above ground, H metres, of a mechanical arm can be modelled by the differential equation

$$\frac{dH}{dt} = \frac{2H\cos 0.2t}{15}$$

where t is the time in seconds since the mechanical arm was switched on. The arm is initially 2 m above the ground.

a) Show that $H = 2e^{\frac{2}{3}\sin 0.2t}$

b) Determine the first time after being switched on that the mechanical arm reaches its maximum height.

2.40 — Question 17. The rate of change of y with respect to x is directly proportional to $\frac{10-x}{y}$. When $x = 5$, both $y = 20$ and the rate of change of y with respect to x is 10. Show that

$$y^2 = 800x - 40x^2 - 2600$$

2.40 — Question 18. A piece of paper has ink dropped onto it. Initially the ink blob has a radius of 4 mm and after 2 seconds has a radius of 16 mm. In a simple model, the rate of increase of the radius of the ink blob is inversely proportional to the square root of the radius.

a) Using this model and defining any variable you use, find an equation linking the radius and the time.

b) Explain why this model may not be valid in the long-term.

2.40 — Question 19. A particle's acceleration, a m s^{-2}, at time t seconds, may be modelled in terms of its velocity v m s^{-1}, as $a = 1.02v^3$, where $v > 0$. The particle's initial velocity is 1 m s^{-1}. Show that

$$v = \frac{5}{\sqrt{25 - 51t}}$$

2.40 — Question 20. A particle's acceleration, a m s^{-2}, at time t seconds, may be modelled in terms of its velocity v m s^{-1}, as $a = -0.02v^2$. The particle's initial velocity is 2 m s^{-1}. Show that

$$v = \frac{50}{t + 25}$$

2.40 — Question 21. A particle's acceleration, a m s^{-2}, at time t seconds, may be modelled in terms of its velocity v m s^{-1}, as $a = -1.05v$. The particle's initial velocity is 6 m s^{-1}. Determine the speed of the particle when $t = 2$, giving your answer to 3 significant figures.

2.40 — Question 22. Dan measures the temperature, T°C, of a freshly made cup of coffee after t minutes. He assumes that the temperature decreases at a rate that is inversely proportional to the square root of coffee's temperature.

a) Construct a differential equation involving T, t and a constant k, where $k > 0$, to model this situation.

b) Explain why this model may not be valid.

2.40 — Question 23. A water tank contains a large amount of water with volume V m^3. There is a small hole at the bottom of the tank such that water flows out at a constant rate of 0.02 m^3 per minute. At time $t = 0$, the volume of the water in the tank is 20 m^3. A tap is turned on so that water flows into the tank at a rate proportional to the volume of the water already in the tank. Show that

$$50\frac{dV}{dt} = kV - 1$$

where k is a positive constant.

2.40 — Question 24. £500 is invested in a savings account. The amount of money in the account after t days is £P. The rate at which P increases is kP, where k is a positive constant.

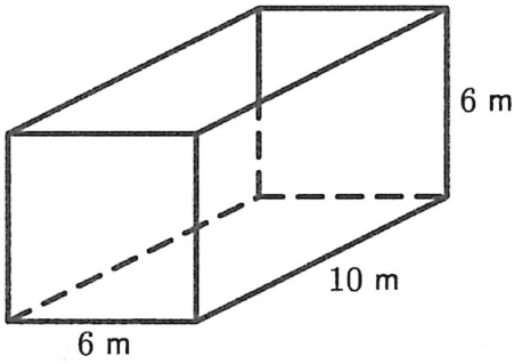
a) Construct a differential equation involving P, t and k.

b) It is given that initially, the rate of increase of the money in the savings account is £0.02 per day. Determine how much money is in the account when $t = 365$

2.40 — Question 25. The rate of increase of bacteria in a petri dish is modelled as being directly proportional to the number of bacteria, N, at any time. When the bacteria are increasing at 5000 a day, there are 60000 bacteria. Write down a differential equation to model this.

2.40 — Question 26. The rate at which a tree's height, h m, increases is known to be directly proportion to the natural logarithm of the tree's height at any time, t years. When the tree is 2 m tall, the rate at which the tree is growing is 0.2 m per year. Write down a differential equation to model this.

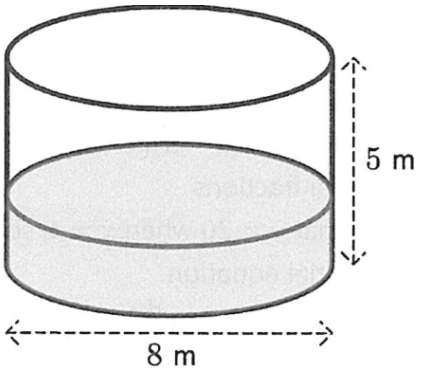
2.40 — Question 27. The diagram below shows a cuboid water tank, which has a width of 6 m, a depth of 10 m and a height of 6 m. The depth of the water in the tank is d metres.



There is a small hole at the bottom of the tank such that water flows out at a rate of $0.05d$ m^3 per minute. A tap is turned on so that water flows into the tank at a rate of 0.1 m^3 per minute. Show that

$$\frac{dV}{dt} = \frac{120 - V}{1200}$$

2.40 — Question 28. The diagram below shows a cylindrical water tank, which has a height of 5 m and a diameter of 8 m. The depth of the water in the tank is d metre time t minutes.



There is a small hole at the bottom of the tank such that water flows out at a rate of $0.2d^2$ m^3 per minute. A tap is turned on so that water flows into the tank at a rate of 0.8 m^3 per minute. Show that

$$\frac{dV}{dt} = \frac{(32\pi - V)(32\pi + V)}{1280\pi^2}$$

2.40 — Question 29. Prerequisite knowledge for Q29–Q46: Connected Rates of Change, Double Angle Formulae, Integration by Substitution, Integration by Parts, Partial Fractions: Solve the following differential equations:

a) $\dfrac{dy}{dx} = \dfrac{(x+2)^3}{y}$

b) $\dfrac{dy}{dx} = y(2x-5)^4$

c) $\dfrac{dy}{dx} = \dfrac{\sqrt{4x+7}}{\sqrt{y}}$

d) $\dfrac{dy}{dx} = \dfrac{(3-x)^{2/3}}{e^y}$

2.40 — Question 30. Solve the following differential equations:

a) $\dfrac{dy}{dx} = \dfrac{xe^x}{y}$

b) $\dfrac{dy}{dx} = xy\sin 10x$

c) $\dfrac{20}{x}\dfrac{dy}{dx} = \dfrac{e^{2x}}{y^2}$

d) $\dfrac{3}{x\ln x}\dfrac{dy}{dx} = \dfrac{5}{\ln y}$

2.40 — Question 31.

a) Express

$$\frac{8x-10}{(x-3)(x+4)}$$

as partial fractions

b) Given that $y = 20$ when $x = 4$, solve the differential equation

$$(x-3)(x+4)\frac{dy}{dx} = 2y(4x-5)$$

writing your answer in the form $y = f(x)$

2.40 — Question 32. Solve the differential equation

$$\frac{dy}{dx} = 8y\cos x \sin^3 x$$

such that $y = 2$ when $x = 0$, writing y in terms of x.

2.40 — Question 33. A spherical balloon is deflating so that at time t seconds, the balloon has radius r cm and volume V cm^3. The volume of the balloon is modelled as decreasing at a constant rate.

a) Using this model, show that

$$\frac{dr}{dt} = -\frac{m}{r^2}$$

where m is a positive constant.

b) It is given that the balloon's initial radius is 30 cm, and after 2 seconds the radius is 29 cm. The volume of the balloon continues to decrease at a constant rate until the balloon is completely deflated. Solve the differential equation, writing your answer in the form $r = f(t)$

2.40 — Question 34. A large cylindrical drum contains a volume of water. The area of the base of the drum is 3600 cm^2. The depth of the water after t seconds is h cm. A tap is turned on and water pours into the drum at a constant rate of 2400 cm^3 s^{-1}. There is a hole at the bottom of the tank such that water flows out at a rate proportional to the square of the height of the water already in the cylinder. Show that at time t seconds, the height of the water in the drum satisfies the differential equation

$$\frac{dh}{dt} = \frac{2}{3} - kh^2$$

where k is a positive constant.

2.40 — Question 35. It is given that $P > 0$ and that

$$\frac{dP}{dt} = \frac{\sin^2 4t}{10P}$$

Given also that $P = 20$ when $t = 0$, show that

$$P = \frac{1}{20}\sqrt{160000 + 40t - 5\sin 8t}$$

2.40 — Question 36.

a) Solve the differential equation

$$\frac{dy}{dx} = \frac{y}{10}(1 - \frac{y}{100})$$

where $0 < y < 100$.

It is also given that $y = 10$ when $x = 0$.

b) Find the value of y as x tends to infinity.

2.40 — Question 37.

a) Express

$$\frac{1}{x(200-x)}$$

as partial fractions.

b) It is given that $\frac{dx}{dt} = \frac{x(200-x)}{1000}$ where $0 < x < 200$ and $t \geq 0$.
It is also given that $x = 100$ when $t = 0$.
Show that

$$x = \frac{200e^{t/5}}{1+e^{t/5}}$$

c) As $t \to \infty$, $x \to k$. Find the value of k.

2.40 — Question 38. A study into the population, P thousands, of a certain type of insect can be modelled by the differential equation

$$15000\frac{dP}{dt} = P(1000-P)$$

where t is the time measured in weeks since the study began. It is given that initially there are 4000 insects.

a) Solve the differential equation to show that

$$P = \frac{4000}{a+be^{-x/15}}$$

where a and b are constants to be found.

b) Hence show that the population will not drop below 1000.

2.40 — Question 39.

a) Use the substitution $u = 2\sqrt{x} - 1$ to show that

$$\int \frac{1}{2\sqrt{x} - 1}\, dx = \sqrt{x} + \frac{1}{2}\ln|1 - 2\sqrt{x}| + c$$

where c is a constant.

b) Hence solve the differential equation

$$\frac{dx}{dt} = \frac{2\sqrt{x} - 1}{50t^{2/5}}$$

given that $x = 0$ when $t = 0$ to show that

$$t = (30\sqrt{x} + 15\ln|1 - 2\sqrt{x}|)^{5/3}$$

c) Find, according to the model, the range of values of x.

2.40 — Question 40. The gradient function of a curve is given by

$$\frac{dy}{dx} = \frac{4xe^{2x}}{\cos 4y}$$

Solve the differential equation to show that

$$y = \frac{1}{4}\arcsin(4e^{2x}(2x - 1) + c)$$

2.40 — Question 41. The gradient function of a curve is given by

$$\frac{dy}{dx} = \frac{(3x-1)^5}{y \sin 4y}$$

Find an equation for the curve in the form $f(y) = g(x)$.

2.40 — Question 42. The gradient function of a curve is given by

$$\frac{dy}{dx} = \frac{6 \cos^2 2x}{y^2 \sin 3y}$$

Find an equation for the curve in the form $f(y) = g(x)$.

2.40 — Question 43. A colony of 2000 insects is introduced to an area. The rate of increase of the number of insects, N, after t weeks is modelled by

$$\frac{dN}{dt} = \frac{9000000 - N^2}{500}$$

Determine the long-term number of insects in the colony.

2.40 — Question 44. A cuboid water tank has base dimensions of 5 m by 3 m. The depth of the water after t seconds is h m. Initially the tank is empty. There is a hole at the bottom of the tank such that water flows out at a rate of $0.1h^2$ m^3 per second. At time $t = 0$, a tap is turned on so that water flows into the tank at the constant rate of 0.4 m^3 per second. Find the time taken for the water to reach a depth of 10 cm.

2.40 — Question 45. Given $y = 1$ when $x = 0$, solve:

$$\frac{dy}{dx} = \frac{y}{8 - 2\sqrt{x}}$$

2.40 — Question 46. The gradient function of a curve is given by the differential equation

$$(3x + 1)^3 \frac{dy}{dx} - 16y^2 = 0$$

Given that $y = 2$ when $x = 1$, show that

$$y = \frac{ax^2 + bx + c}{px^2 + qx + r}$$

where a, b, c, p, q, r are integers to be found.